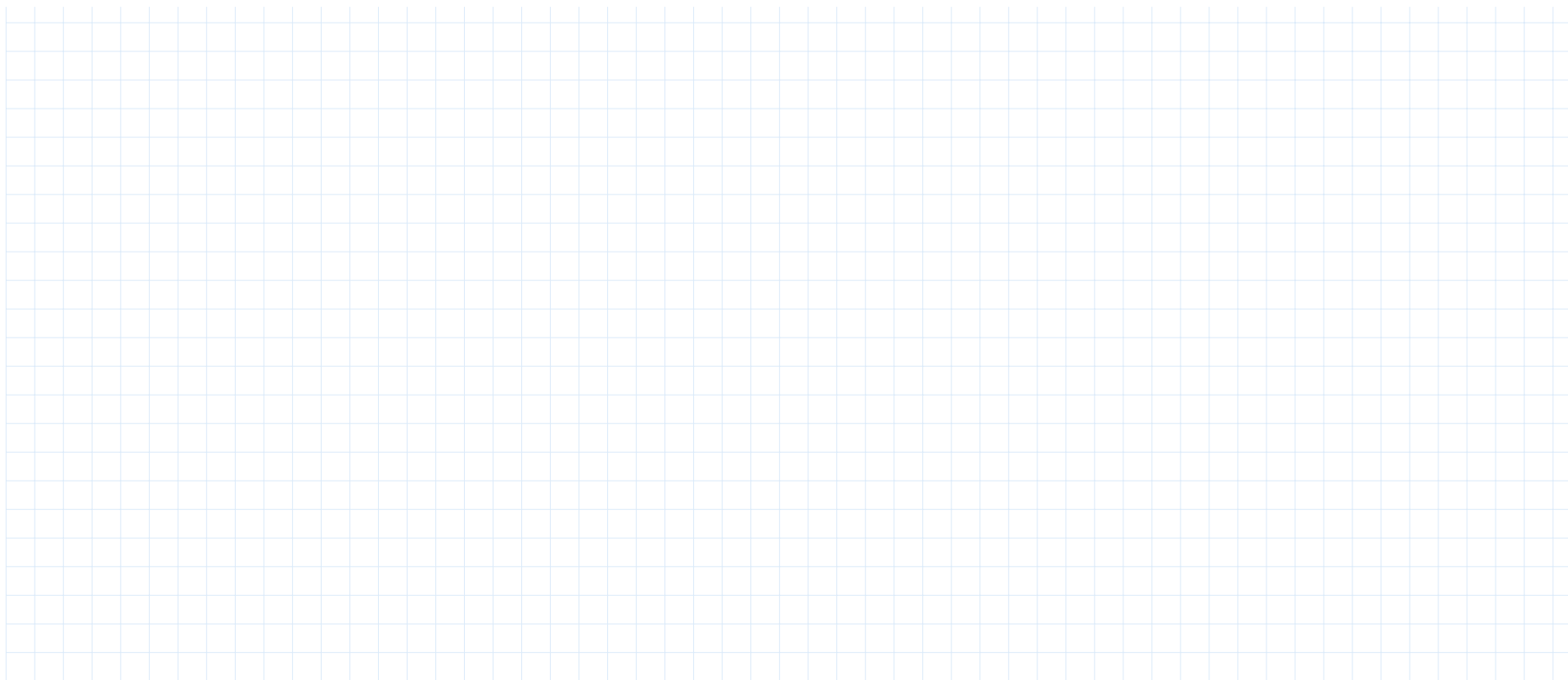


The tangent graph

Since $\tan x = \frac{\sin x}{\cos x}$, tangent is undefined where cosine is zero.

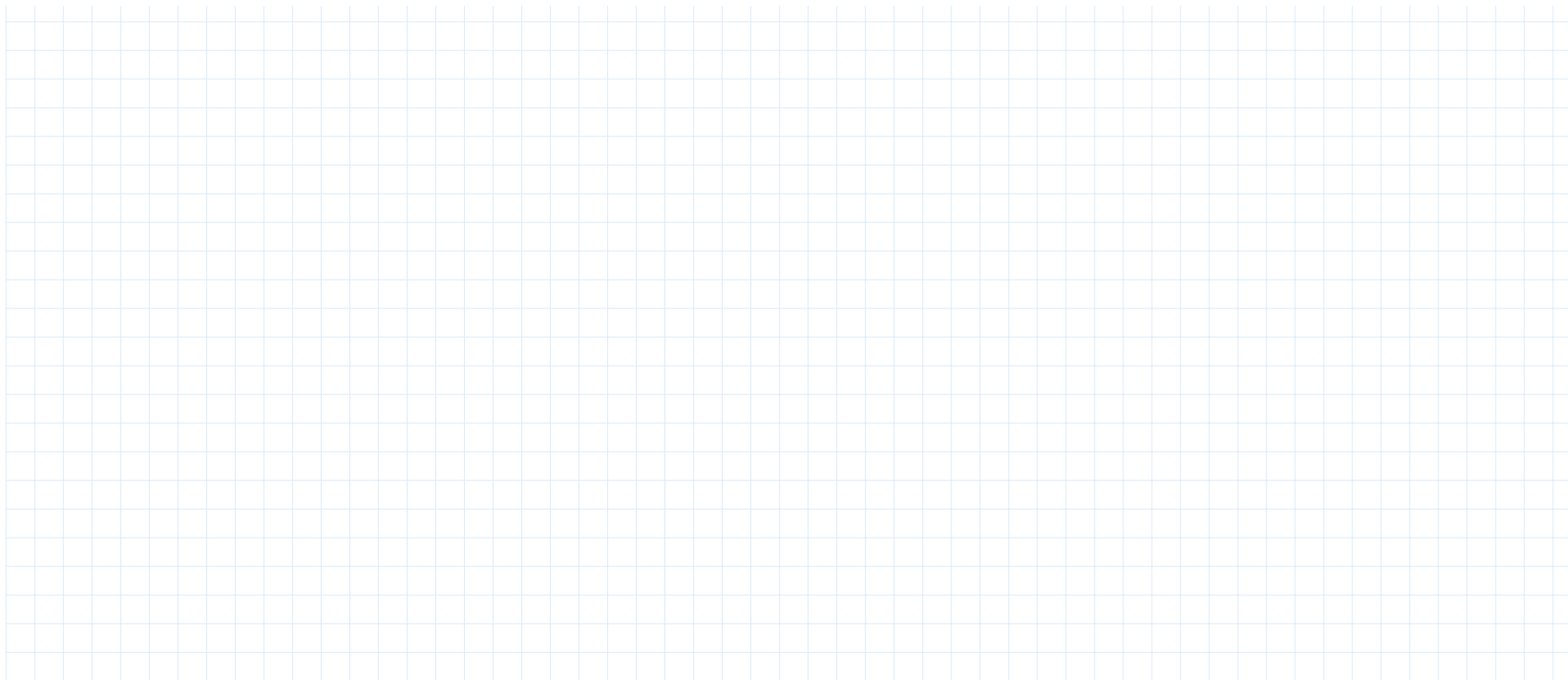
Example. Sketch one cycle of $y = \tan x$ on $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Mark the asymptotes and three useful points. State the domain, range, and period of the full function.



The cotangent graph

Since $\cot x = \frac{\cos x}{\sin x}$, cotangent is undefined where sine is zero.

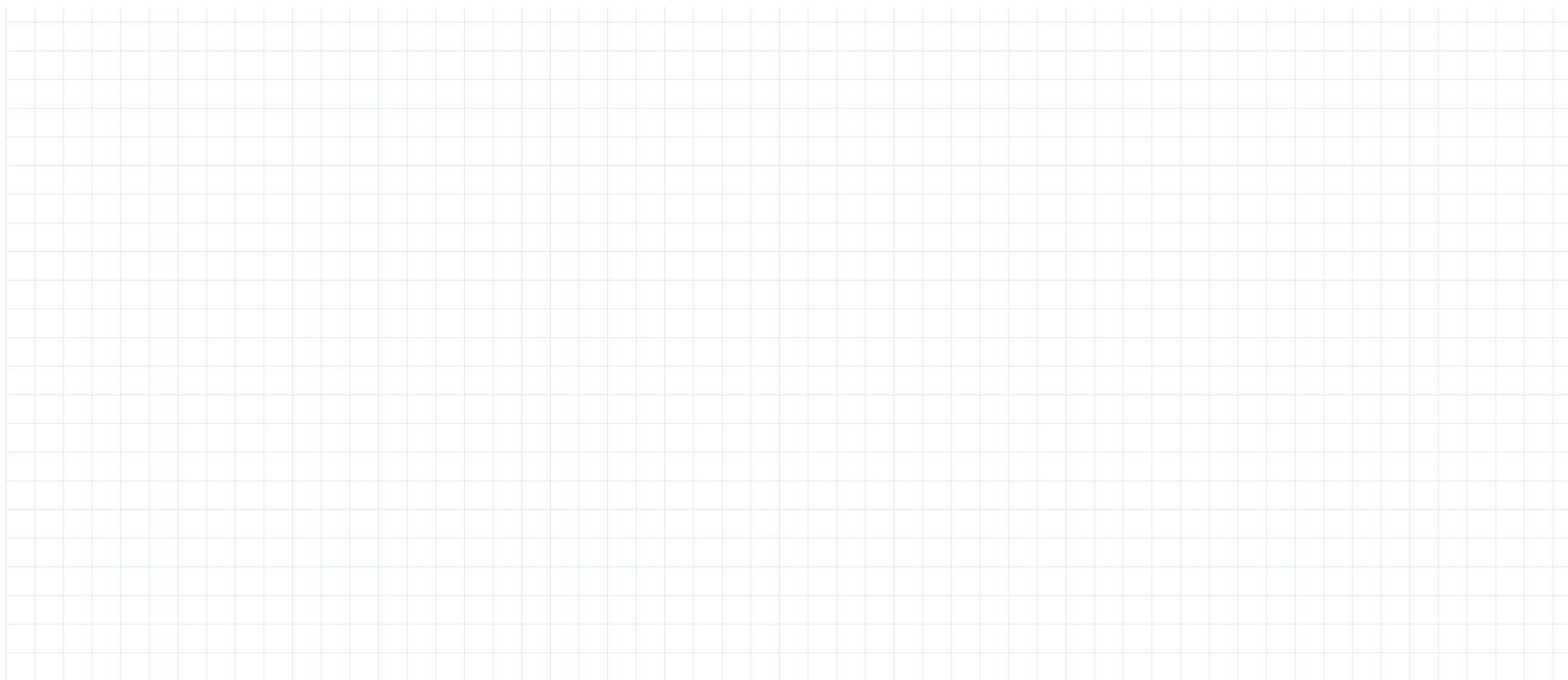
Example. Sketch one cycle of $y = \cot x$ on $0 < x < \pi$. Mark its asymptotes, intercept, and two additional points. State the full domain, range, and period.



The secant graph

Since $\sec x = \frac{1}{\cos x}$, zeros of cosine become vertical asymptotes of secant.

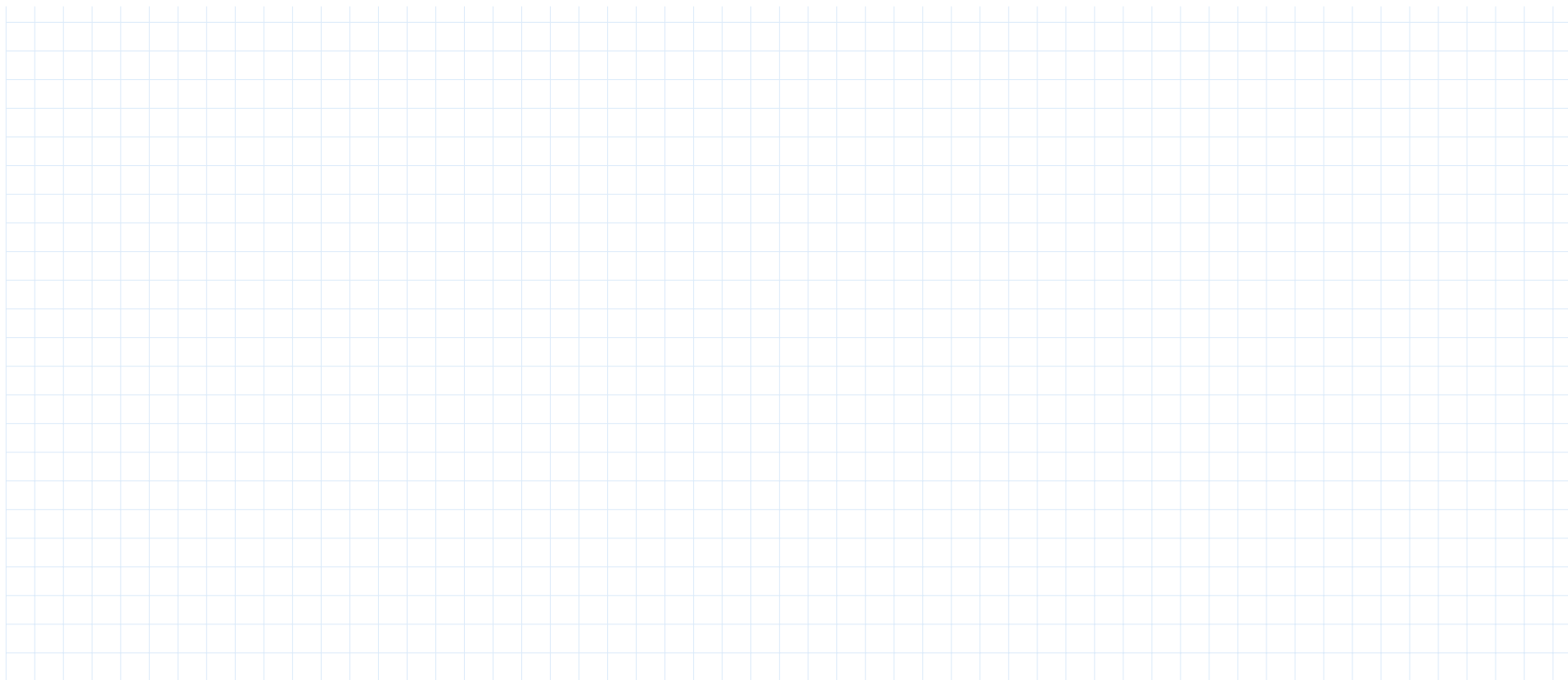
Example. Sketch $y = \sec x$ on $0 \leq x \leq 2\pi$. Mark the asymptotes and vertices. State the full domain, range, and period.



The cosecant graph

Since $\csc x = \frac{1}{\sin x}$, zeros of sine become vertical asymptotes of cosecant.

Example. Sketch $y = \csc x$ on $0 < x < 2\pi$. Mark the asymptotes and vertices. State the full domain, range, and period.



Transforming tangent

For $y = A \tan(Bx) + D$, the period is $\frac{\pi}{|B|}$. The coefficient A stretches or reflects the graph; it is not an amplitude.

Example. Sketch one cycle of $y = -2 \tan\left(\frac{x}{2}\right) + 1$. Find its period, central asymptotes, full domain, and range.

